Horizontal Investments: Information and Competition

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Motivation

Vertical investments, and the frictions that affect them, have been widely studied.

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How do market frictions affect horizontal investment?

Model Overview

Agents with differing vertical quality make a horizontal investment decision.

Within a horizontal sector, agents match with heterogeneous positions.

- Surplus is created, depending on the matched pair.
- This surplus is divided between the agent and position in the pair.

I explore the impact of two frictions:

(i) **Information frictions**: whether agents know their quality at the time of their horizontal investment decision.

(ii) **Competition frictions**: whether agents can compete for positions by accepting a smaller share of the created surplus.

Main Result

	No Quality Uncertainty	Quality Uncertainty
Competitive Market	No Inefficiency	No Inefficiency
Noncompetitive Market	No Inefficiency	**“Allocative” Inefficiency**

Applications

Apply the framework to three applications:

	Horizontal	Agents (Quality)	Positions (Attribute)
Workers	Sector	Workers (Productivity)	Firms (Productivity)
Innovation	Field	Researchers (Success Time)	Innovation (Value)
Production	Genre	Producers (Quality of Good)	Demand (Quantity)

Motivating Example

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There are two agents, and two sectors, each with one position.

- Agents quality are distributed according to $v_i \sim \text{Unif}[1, 2]$.
- Sector 1's position has attribute $z_1 > \frac{1}{2}$, Sector 2's has attribute $z_2 = 1 - z_1$.
- If agent i matches with the position in sector j, surplus $v_i \cdot z_j$ is created.

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Quality then becomes public, agents match with positions, and divide surplus.

- In competitive markets, agents propose a surplus division to their sector's position. A position accepts its best offer.
- In noncompetitive markets, matched agents receives surplus proportional to the created surplus, so the best agent is matched.

Proposition

(i) If quality is public before agents choose sectors, the equilibrium is efficient;

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(ii) If markets are competitive, there exists an efficient equilibrium; but

- If both agents enter sector 1, the worse agent $-i$ receives 0, the better agent i receives

$$(v_i - v_{-i}) \cdot z_1$$

- Agents internalize their *displacement* of the other.
  - This leads to efficiency.

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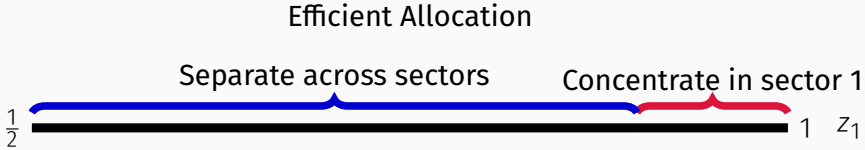


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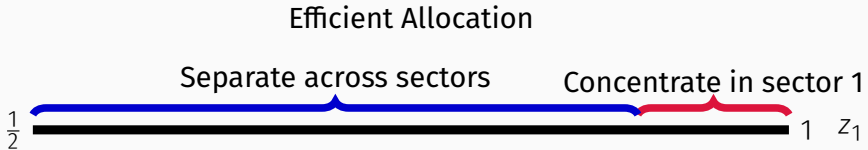
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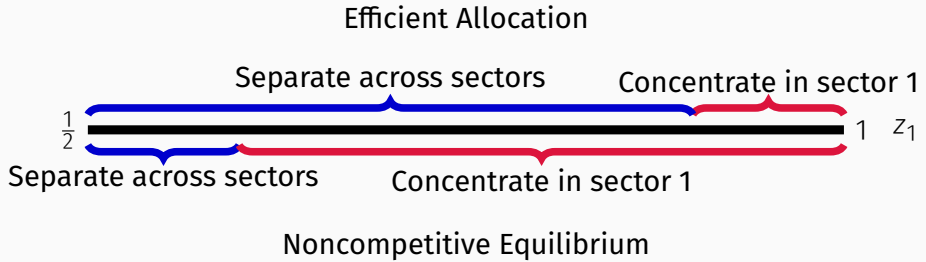
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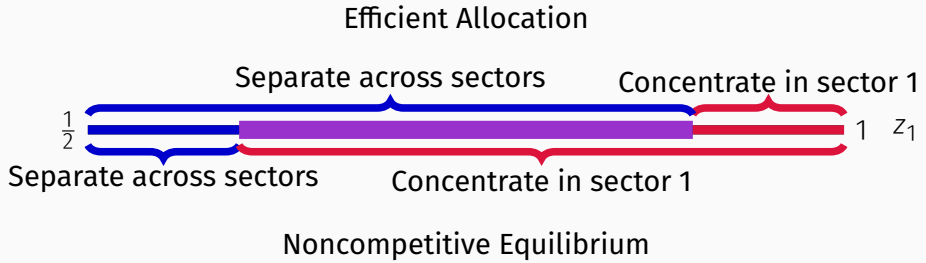


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Model

Agents and Sectors

Two-sided market, with one side divided into discrete sectors.

- Continuum of **agents** $i \in [0,1]$, each with quality $v_i \sim F$.
- Sectors $\mathcal{S} = \{1, ..., S\}$, each with a mass of **positions**.
- In sector s, position $j_s \in [0,1]$ has attribute $z_{j_s} \sim G_s$.

An agent commits to a sector and is matched with a position in that sector.

- Agent i and position j_s produce and divide surplus $\sigma(v_i, z_{j_s})$.

Assumption

(i) F is full-support on $[0, \overline{v})$ for $\overline{v} \in (0, \infty]$, with density f;

(ii) G_s on $Z_s := [0, \overline{z}_s]$ for $\overline{z}_s \in (0,1]$ has density g_s bounded from zero; and

(iii) $\sigma(v,z)$ is strictly supermodular with $\sigma(0,z) = \sigma(v,0) = 0$.

Agents' Types

An agent's type $\theta \in \Delta([0,1])$ is their belief about their rank within the population.

- $\Theta \subseteq \Delta([0,1])$ the set of agent types.
- $\tau \in \Delta\Theta$ the distribution of types.

Cases:

- $\Theta_C = \{\delta_x : x \in [0,1]\}$: complete information.
- $\Theta_N = \{U[0,1]\}$: no information.
- Agents have incomplete private information about their rank.

Consistency requirement on τ: $\mathbb{E}_\tau[\theta] = U[0,1]$.

Assumption

For today: (i) Θ is finite and affinely independent, or (ii) $\Theta = \Theta_C$.

Agents' Strategy

A population strategy for agents is a mapping

$$\Phi : \Theta \to \Delta(\mathcal{S})$$

This strategy induces a rank-measure of agents Ψ_s in each sector s:

$$\Psi_s = \int \Phi_s(\theta) \cdot \theta \cdot d\pi(\theta),$$

and a distribution of values F_s:

$$F_s(v) \coloneqq \underbrace{\Psi_s([0, F(v)])}_{\text{Conversion from rank to value}} + \underbrace{(1 - \Psi_s([0, 1]))}_{\text{Remaining mass}}$$

Let $V_s \coloneqq \operatorname{supp}[F_s]$.

Within-Sector Matching

Given Ψ_S, agents' qualities are realized, they match with positions.

Definition

A *matching function* $M : Z_S \to V_S$ is a measurable function that is measure-preserving with respect to G_S, F_S.

Division of Surplus

After agents match with positions, created surplus is divided:

Definition

- Given Ψ_s, an agent-surplus profile is a function $\sigma_{a,s} : V_s \to \mathbb{R}_+$.
- A sector s position-surplus profile is a function $\sigma_{p,s} : Z_s \to \mathbb{R}_+$.

If a matching M is feasible $\sigma_{a,s}(M(z)) + \sigma_{p,s}(z) \leq \sigma(M(z), z)$.

There are two possibilities for how this division is decided:

- Competitively: Agents and positions compete through surplus offers; or
- Noncompetitively: There exists an exogenous division of surplus.

Definition

A sector s *competitive equilibrium* given Ψ_s is a matching function M and surplus profiles $\sigma_{a,s}, \sigma_{p,s}$ such that

$$\sigma_{p,s}(z) = \sigma(M(z), z) - \sigma_{a,s}(M(z)) \geq \sigma(v, z) - \sigma_{a,s}(v), \quad \text{for all } z, v \quad \text{(Stab)}$$

$$\sigma_{a,s}(v) \geq 0, \quad \text{for all } v \quad (\text{IR}_a)$$

$$\sigma_{p,s}(z) \geq 0, \quad \text{for all } z \quad (\text{IR}_p)$$

Division of Surplus: Competitive Markets

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Lemma

Given Ψ_s, any sector s competitive equilibrium features positive assortative matching (PAM): $M(z) = F_s^{-1}(G_s(z))$.

Division of Surplus: Noncompetitive Markets

Noncompetitive markets: an exogenous (sector-independent) division rule.

Definition

A division of surplus $\sigma_a, \sigma_p : \mathbb{R}^2_+ \to \mathbb{R}$ is *admissible* if:

(i) $0 \leq \sigma_a(v, z), \sigma_p(v, z)$;

(ii) $\sigma_a(v, z) + \sigma_p(v, z) \leq \sigma(v, z)$; and

(iii) $\sigma_a(v, z)$ is increasing in z, and $\sigma_p(v, z)$ is increasing in v.

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(iii) $\sigma_a(v,z)$ is increasing in z, and $\sigma_p(v,z)$ is increasing in v.

Lemma

Given admissible σ_a, σ_p, for all Ψ_s, PAM is stable.
Under PAM, agent v's surplus in sector s is $\sigma_{a,s}(v) := \sigma_a\left(v, G_s^{-1}(F_s(v))\right)$.

Equilibrium

Definition

An equilibrium of the game is a strategy $\Phi : \Theta \to \Delta(\mathcal{S})$, distributions of agents $(\Psi_1, ..., \Psi_S)$, and values for agents $\left(\sigma_{a,s}(v)\right)_{s \in S}$ such that

(i) For all s and v, $\sigma_{a,s}(v)$ represent equilibrium payoffs in the sector-subgame given Ψ_S;

(ii) If $\Phi_s(\theta) > 0$, then $s \in \text{argmax}_{s'} \mathbb{E}_\theta \left[\sigma_{a,s'}(F^{-1}(r))\right]$; and

(iii) $\Psi_s = \int \Phi_s(\theta) \cdot \theta \cdot d\tau(\theta)$.

Main Result

Within-Sector Efficiency

Regardless of the market structure, the resulting matching is efficient:

Lemma

Given an allocation of producers Ψ_S, there is a unique competitive sector subgame equilibrium. It is efficient. If the market is not competitive, there is an efficient stable matching.

- This allows me to focus exclusively on Ψ_S.

Allocative (In)Efficiency

Definition

Given Ψ_s, *aggregate surplus* in sector s is: $W_s(\Psi_s) \coloneqq \int_0^1 \sigma\left(F_s^{-1}(r), G_s^{-1}(r)\right) dr$.

Given agents' information, the feasible $\Psi = (\Psi_1, ..., \Psi_S)$ are

$$\mathcal{F}(\tau) \coloneqq \left\{ \Psi : \exists \Phi : \Theta \to \Delta(\mathcal{S}) \text{ measurable}, \Psi_s = \int \Phi_s(\theta) \cdot \theta \cdot d\tau(\theta) \right\}$$

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Definition

An allocation of producers $\Psi^* = \left(\Psi_1^*, ..., \Psi_S^*\right)$ is *allocatively efficient* (*given* Θ, τ) if

$$\Psi^* \in \underset{\Psi \in \mathcal{F}(\tau)}{\operatorname{argmax}} \sum_{s \in \mathcal{S}} W_s(\Psi_s)$$

I say that Ψ^* is *allocatively inefficient* if it is not allocatively efficient.

Main Result: Both Frictions $\iff$ Allocative Inefficiency

Theorem

Fix σ, admissible σ_a, σ_p, and (Θ, τ).

(i) If agents have perfect information, $\Theta = \Theta_C$, the unique equilibrium is efficient;

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(i) If agents have perfect information, $\Theta = \Theta_C$, the unique equilibrium is efficient;

(ii) If markets are competitive, the unique equilibrium is efficient; but

(iii) If agents have no private information $\Theta = \Theta_N$ and markets are not competitive, generically* the unique equilibrium is inefficient.

*Genericity is with respect to the distributions of agents and positions (F, G).

Extending to Private information: Need mixing from some type

Proposition

Suppose that $\Theta \neq \Theta_C$ and markets are not competitive.

(i) The set of (F, G) for which every equilibrium is inefficient is open; and

(ii) If the efficient allocation features mixing from some type θ, then there exists arbitrarily small perturbations to (F, G) that make every equilibrium inefficient.

- If types have strict preferences over sectors, equilibrium in the market without competition may be efficient.
  - However, congestion pushes types towards mixing.
- Condition (ii) is generic if $\Theta = \Theta_N$.

Proof Sketch: (i) Complete information implies efficiency

If $\Theta = \Theta_C$, then any allocation Ψ is feasible.

- Efficiency is equivalent to, *across sectors*, PAM is satisfied.

Noncompetitive markets:

- The best agents will go to the sectors with the best positions $\Rightarrow$ PAM.

Competitive markets:

- Less obvious: division profiles could differ across sectors.
- In equilibrium, competitive forces are the same across sectors $\Rightarrow$ best agents go to the sectors with the best positions.

Proof Sketch: (ii) Competition implies efficiency

Lemma

$W_S(\Psi_S)$ is strictly concave, as a function of Ψ_S.

Corollary

There exists a unique efficient Ψ. It is identified from first order conditions.

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There exists a unique efficient Ψ. It is identified from first order conditions.

The bulk of the proof is showing that, for all allocations Ψ_S,

$$\frac{d}{d\epsilon} W_S\left(\Psi_S + \epsilon\delta_{F(v)}\right)\Big|_{\epsilon=0} = \sigma_{a,S}(v;\Psi_S)$$

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$$\frac{d}{d\epsilon} W_S\left(\Psi_S + \epsilon\theta\right)\Big|_{\epsilon=0} = \int \sigma_{a,S}(F^{-1}(r); \Psi_S) d\theta(r)$$

At an optimal allocation types θ go to the sectors with largest benefit. These are exactly those with the largest competitive surplus.

Proof Sketch: (iii) Noncompetitive Efficiency is Knife-Edge

Suppose that $\Theta \neq \Theta_C$, and some type θ is mixing in equilibrium.

- In competitive equilibrium, divisions rely on the *entire* distribution of assignments in a sector.
- In a noncompetitive equilibrium, agent v's payoff depends only on his match.

Proof Sketch: (iii) Noncompetitive Efficiency is Knife-Edge

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- In competitive equilibrium, divisions rely on the *entire* distribution of assignments in a sector.
- In a noncompetitive equilibrium, agent v's payoff depends only on his match.

As type θ is mixing, perturbing his payoffs in sector s changes his strategy.

- Perturb G_s to keep *all* agents' competitive payoffs in sector s the same, but
- change the noncompetitive payoff for type θ in sector s.
- Given Θ, may need to also perturb F to ensure that this is possible.

Applications

Comparative Statics: Concentration of Environments

Comparative statics depend on the specific application.

Definition

- An environment: $E = \left[\left(\sigma, \sigma_a, \sigma_p\right), \left(\Theta, \tau, F\right)\right]$.
- A complete environment: (E, G) an environment E with $G = (G_1, ..., G_S)$.

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First, study how *concentrated* are environments:

Definition

- Given (E, G), let $A_{E,G} \coloneqq \{s \in \mathcal{S} : \Psi_s \neq 0, \text{ some eq'm}\}$ denote its *active sectors*.
- E is *more concentrated* than E' if for all G, $A_{E,G} \subseteq A_{E',G}$.

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Definition

- Given (E, G), $\theta \in \Theta_E$, let $A_{E,G}(\theta) \coloneqq \{s \in \mathcal{S} : \Phi_s(\theta) > 0, \text{ some eq'm}\}$ denote θ's *active sectors*.
- Fix E, E' with $\theta \in \Theta_E = \Theta_{E'}$, $\tau_E = \tau_{E'}$:
  θ *concentrates more* in E than E' if, for all G, $A_{E,G}(\theta) \subseteq A_{E',G}(\theta)$.

Comparative Statics: Concentration of Sectors

Second, study concentration *within* sectors:

Definition

G is *more concentrated* than G' if G' is a mean-preserving spread of G.

When agents have no private information ($\Theta = \Theta_N$), the unique equilibrium just specifies the mass of agents Φ_s in each sector s.

Definition

Fix $E = \left[\left(\sigma, \sigma_a, \sigma_p\right), \left(\Theta_N, \delta_U, F\right)\right]$. Concentrated sectors are *preferred* in E if, for all G_1, G_1' with G_1 more concentrated than G_1' and all G,

$$\Phi_1\Big[\left(E, \left(G_1, G\right)\right)\Big] \geq \Phi_1\Big[\left(E, \left(G_1', G\right)\right)\Big]$$

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Definition

Fix $E = \left[\left(\sigma, \sigma_a, \sigma_p\right), \left(\Theta_N, \delta_U, F\right)\right]$. Concentrated sectors are *eventually preferred* in E if, for all G_1, G_1' with G_1 more concentrated than G_1' there exists $\overline{z}, \delta > 0$ such that for all G with $G_s(\overline{z}) \leq \delta$ for all $G_s \in G$,

$$\Phi_1\Big[\left(E, \left(G_1, G\right)\right)\Big] \geq \Phi_1\Big[\left(E, \left(G_1', G\right)\right)\Big]$$

Worker Sorting

Worker Sorting: Setup

A one-to-one mapping to our framework:

- Agents: Workers, with productivity $v_i \sim F$.
- Positions: Firms, with idiosyncratic productivity $z_{j_s} \sim G_s$ in sector s.

Together, a v worker and z firm produce output $\sigma(v, z) = v \cdot z$.

- If wage frictions exist, a worker is paid proportionately to their output: $\sigma_\alpha(v, z) = \alpha\sigma(v, z)$.

Assumption

F has increasing hazard rates: $\frac{f(v)}{1-F(v)}$ is increasing.

Worker Sorting: Concentration with Grade Inflation

Definition

Θ features "grade inflation" if there exists an $\bar{r}$ such that

$$\Theta = \{\delta_x : x \in [0, \bar{r})\} \bigcup \{U([\bar{r}, 1])\}$$

Proposition

Given a Θ with grade inflation, top workers concentrate more in the noncompetitive equilibrium than in the efficient allocation.

- In noncompetitive markets, top workers do not internalize their displacement of other workers.
- This leads them to compete too heavily for positions in attractive sectors.

Worker Sorting: Workers prefer riskier sectors than is optimal

Now, workers have no information: $\Theta = \Theta_N$.

Proposition

(i) If markets are noncompetitive, less concentrated sectors are preferred; but

(ii) If markets are competitive, concentrated sectors are eventually preferred.

When markets are noncompetitive, workers prefer *risky* sectors: those with some very attractive positions and many unattractive positions. Instead, a planner prefers many similar positions, since returns are high.

Innovation

Innovation: Model I

There is a unit mass of researchers $i \in [0, 1]$.

- Researcher i achieves an innovation at rate λ, independently;
- Researcher i chooses a discrete "sub-field" in which to work.

There are $\mathcal{S}$ *sub-fields*.

- In sub-field s there is a distribution of innovations of varying quality $z \sim G_s$.
- Think of as the value of the innovation.

Time is discounted by society at rate r, and by researchers at rate δ, with $\delta \geq r$.

Innovation: Model II

Researchers commit to a sub-field, and then compete with other researchers in their sub-field to achieve innovations.

- The first researcher to achieve an innovation z at time t receives $ze^{-\delta t}$. This captures the benefit from patenting.
- The social benefit is ze^{-rt}.

Consider two regimes. After his first achieved innovation, researchers either

- Exit (develop a research agenda); or
- Remain (sequential research).

Innovation: Mapping to Framework

Agents: Researchers, with qualities e^{-rt}.

Positions: Innovations, with value z.

Surplus generation function $\sigma(e^{-rt}, z) = z \cdot e^{-rt}$, with $\sigma_a(e^{-rt}, z) = z \cdot e^{-\delta t}$.

Information $\Theta = \Theta_N$: arrivals are random.

The last piece is the distribution F. This depends on the regime:

Lemma

- In *Exit*: $e^{-rt} \sim \text{Beta}\left(\frac{\lambda}{r}, 1\right)$;
- In *Remain*: e^{-rt} follows an improper distribution on $(0, 1]$, with density $\frac{\lambda}{r} \cdot \frac{1}{x}$.

In "Remain", all projects are eventually completed, so the distribution is improper.

Innovation: Concentration Results

Proposition

Suppose that $\delta \geq r$. In both regimes:

(i) Researchers are more concentrated than is socially optimal;

(ii) As the rate of discovery λ increases, concentration decreases; and

(iii) As researchers become more impatient (δ increases), concentration increases.

- As in the worker application, market frictions cause agents to concentrate too heavily.
- Impatience (e.g., short funding windows) exacerbates this problem.

Innovation: Researchers prefer risky fields

Proposition

Let $\delta = r$. In both regimes,

(i) researchers prefer less concentrated (risky) sub-fields;

(ii) if markets were competitive and $\frac{\lambda}{r} \geq 1$, then concentrated subfields are eventually preferred; but

(iii) If $\frac{\lambda}{r} < 1$, concentrated subfields are not eventually preferred.

The "speed" of innovation matters in competitive markets:

- If $\frac{\lambda}{r} < 1$, innovations are "slow", and so later innovations hold little value.
- The "most preferred" sectors (subject to average innovation) will equalize value across the first $\frac{\lambda}{r}$ mass of projects, and set the value of others to zero.

Innovation: Comparison of Regimes

Proposition

(i) Researchers concentrate more in *Exit* than in *Remain*; but

(ii) In the efficient allocations, it is ambiguous there is more concentration in *Exit* than in *Remain*.

- In *Exit*, competition forces accumulate more slowly: if a researcher is unlucky, competitors will have already left.
- In an efficient market there are two forces. In *Remain*
  - more projects are solved per researcher $\Rightarrow$ spread out; but
  - an additional researcher can finish multiple projects $\Rightarrow$ concentrate.

  Which force wins depends on G.

Content Production

Content Production: Model I

There is a unit mass of producers.

- Each producer produces a good of random value $v \sim F$.

There are S "genres" ("styles") of goods.

- Producers choose in which genre they produce, but do not know their quality ex-ante: $\Theta = \Theta_N$: "Nobody Knows Anything."

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There is a unit mass of heterogeneous consumers with types $\xi \in \Xi$.

- Type $\xi = (t_1, ..., t_S)$ can consume up to $t_s \in [0, 1]$ mass of content in genre s.
- Let $T_s(t)$ is the mass of consumers willing to consume at most t units of content in genre s.
  - Assume that T_s is strictly increasing and continuous (may have an atom at zero).

Content Production: Model II

Consumer of type $\xi = (t_1, ..., t_S)$ receives utility proportional to the quality of goods that he consumes up to his capacity constraint, net their cost.

- He consumes the t_s goods in genre s that give the highest net utility $v - p_s(v)$.

Producers maximize profits, anticipating prices and demand in each genre.

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Today: consider two pricing cases.

- Competitive Pricing: After qualities are realized, producers set prices in each genre to maximize demand $p_s(v)$;
- "Streaming": Consumers do not pay per good, producers aim to maximize demand.

Content Production: Mapping to Framework

Agents: Producers with quality v.

Positions: Consumer demand, with quantity z.

- The distribution of demand in genre s is $G_s(z) \coloneqq T_s^{-1}(z)$.

Surplus generation function: $\sigma(v, z) = v \cdot z$.

Surplus division $\sigma_a(v, z)$ depends on the pricing scheme:

- Competitive pricing: determined in equilibrium.
- *Streaming* pricing: $\sigma_a(v, z) = z$.

Information $\Theta = \Theta_N$: producers do not know their quality.

Lemma

There exists a unique equilibrium of the competitive pricing sector sub-game that corresponds to the unique competitive equilibrium in the general model.

This isn't obvious.

- Producers are the only ones able to make offers, and they must choose a per-unit price.
- Consumers also decide consumption not at the z-level, but individually.
- The assumption that T_S has no atoms (except at zero) is important.

Content Production: Concentration depends on tail-weight

Definition

$\frac{f(v)}{1-F(v)}$ is the hazard rate of F.

Proposition

(i) If F has *increasing* hazard rates, the *Streaming* equilibrium is more concentrated than the efficient allocation; but

(ii) If F has *decreasing* hazard rates, the *Streaming* equilibrium is less concentrated than the efficient allocation.

Content Production: Homogeneous vs. Heterogeneous Consumer Demand

Lemma

If T is more concentrated than T', then the induced G is *less* concentrated than G'.

Comparative statics on T correspond to the "reverse" comparative statics on G.

Proposition

- In the *Streaming* setting, producers prefer homogeneous-demand genres; but
- If F has IHR, efficiently, genres with more heterogeneous consumer demand are eventually preferred.

Content Production: Other Results

Other comparative statics:

- Quality-dependent pricing.
- Quantity-dependent pricing.
- Changes to the value distribution.

I also study how other market structures interact with these inefficiencies:

- A monopolist subject to pricing constraints.
- A monopolist using genre-specific subscriptions.
- A “recommendation system” that connects producers to consumers.

Future Directions

- Tighter results for the general (Θ, τ) inefficiency case.
  - If Θ is rich-enough, the results should go through.
- Other types of comparative statics , possibly for general production functions σ and information structures (Θ, τ).
- General mechanisms for determining surplus divisions.
  - Currently, just the extremes of competitive markets and deterministic divisions. Would like to study intermediate cases.
- Other possible applications?

Related Literature

Matching markets, competitive equilibrium:

- Koopmans and Beckman (1957), Terviö (2008), Kolotilin and Wolitsky (2026)

Worker Sorting:

- Lindenlaub (2017), Lise and Postel-Vinay (2020), Alon and Fershtman (2025)

Innovation:

- Bhattacharya and Mookherjee (1986), Bryan and Lemus (2017), Hopenhayn and Squintani (2021)

Content Production:

- Aguiar and Waldfogel (2018), Gambato (2025), Chen, Li, and Preuss (2025)

Conclusion

I develop a model to understand how information and competition frictions affect *horizontal* investment decisions.

- I show that the uncertainty, together with a lack of competitive pricing, leads to inefficient sorting.

I study three different applications of the framework:

(i) **Worker sorting**: Wage frictions lead to too much concentration for workers.

(ii) **Innovation**: Researchers concentrate and over-value "risky" fields.

(iii) **Content markets**: Direction of inefficiency depends on the quality distribution's tails.